



Complex Numbers · Worksheet 1

Core Pure 1 Chapter 1: Complex Numbers. Answer all questions. Show your working clearly. Total: 81 marks. Time: 100 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 1. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Remember that $i^2 = -1$. Give answers in the form $a + bi$, where a and b are real, and in exact form unless told otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Arithmetic with complex numbers	19	
B Conjugates and division	19	
C Solving quadratic equations with complex roots	18	
D Solving cubics and quartics with a given complex root	20	
Roar-level challenge	5	
Total	81	

Section A Arithmetic with complex numbers

19 marks

1. Simplify

[2]

(a) i^{11} ,

(b) $\sqrt{-49}$.

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2. Express $(3 - 2i)(1 + 4i)$ in the form $a + bi$.

[2]

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3. Given that $z = 2 - i$, find z^2 and z^3 , each in the form $a + bi$.

[3]

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4. Find the real numbers a and b such that $(a + bi)^2 = 5 - 12i$.

[4]

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5. The complex number w is given by $w = 1 + 2i$.

(a) Express w^2 in the form $a + bi$. [2]

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(b) Given that $w^2 + pw + q = 0$, where p and q are real, find the value of p and the value of q . [3]

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(c) Find w^4 in the form $a + bi$, and hence find the real constant k such that $w^4 + kw^2$ is real. [3]

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(Total for Question 5 is 8 marks)



Section B Conjugates and division

19 marks

6. Given that $z = 5 - 3i$, write down z^* and find the value of zz^* . [2]

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7. Express $\frac{3 + 4i}{1 - 2i}$ in the form $a + bi$. [2]

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8. Solve the equation $(1 + i)z = 3 + 5i$, giving z in the form $a + bi$. [3]

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9. Find the complex number z that satisfies the equation [4]

$$z + 2iz^* = 7 + 8i.$$

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10. The complex numbers z_1 and z_2 are given by $z_1 = 2 + 3i$ and $z_2 = 1 - i$.

(a) Find $z_1 z_2$ in the form $a + bi$.

[2]

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(b) Express $\frac{z_1}{z_2}$ in the form $a + bi$.

[3]

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(c) Given that $\frac{z_1 + \lambda}{z_2}$ is purely imaginary, where λ is real, find the value of λ .

[3]

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(Total for Question 10 is 8 marks)



Section C Solving quadratic equations with complex roots

18 marks

11. Solve the equation $z^2 + 16 = 0$. [2]

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12. Solve the equation $z^2 + 6z + 13 = 0$. [2]

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13. Solve the equation $2z^2 - 2z + 5 = 0$, giving your answers in the form $a + bi$. [3]

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14. Given that $3 - 4i$ is a root of the equation $z^2 + bz + c = 0$, where b and c are real constants, find the value of b and the value of c . [4]

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15. The quadratic equation $z^2 - 4z + k = 0$, where k is a real constant, has roots z_1 and z_2 .

(a) Find the range of values of k for which z_1 and z_2 are not real. [2]

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(b) Given that $k = 13$, solve the equation. [3]

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(c) Hence show that $z_1^2 + z_2^2 = -10$. [2]

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(Total for Question 15 is 7 marks)

**Section D Solving cubics and quartics with a given complex root**

20 marks

- 16.** The cubic equation $f(z) = 0$ has real coefficients, and $2 + i$ is a root. **[2]**

Write down another root of the equation, and show that $z^2 - 4z + 5$ is a factor of $f(z)$.

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- 17.** Show that $z = 1$ is a root of $z^3 - z^2 + 4z - 4 = 0$, and hence solve the equation. **[3]**

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- 18.** Given that $1 + 2i$ is a root of the equation $z^3 - 5z^2 + 11z - 15 = 0$, find the other two roots. **[3]**

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- 19.** Given that $3i$ is a root of the equation **[4]**

$$z^4 - 2z^3 + 14z^2 - 18z + 45 = 0,$$

solve the equation completely.

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20.

$$f(z) = z^4 - 4z^3 + 17z^2 - 16z + 52$$

Given that $2 - 3i$ is a root of the equation $f(z) = 0$,

(a) write down another complex root of the equation, [1]

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(b) show that $z^2 - 4z + 13$ is a factor of $f(z)$, [3]

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(c) hence solve the equation $f(z) = 0$ completely. [4]

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(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. Find the two square roots of $-5 + 12i$, giving your answers in the form $a + bi$. [5]
Hence solve the equation $z^2 - (4 - i)z + 5 - 5i = 0$.

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END OF WORKSHEET

Total: 81 marks